Chapter 6

Upholding Responsible AI Standards in GenAI Use

IN THIS CHAPTER

» Figuring out how to make GenAI-generated content original and authentic

» Ensuring GenAI accuracy

» Exploring ethical standards

» Understanding the responsible AI movement

This chapter helps you navigate the exciting yet challenging landscape of GenAI without sacrificing ethical standards and quality. By addressing these critical areas, you can harness the power of GenAI to help create a future that's not only technologically advanced but also grounded in the principles of ethics and accountability.

Achieving Originality and Excellence in GenAI-Generated Content

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generated with GenAI is boundless. Yet, with this remarkable technology comes the responsibility to ensure that the content it generates is both original and of suitable quality.

Originality in GenAI-generated content isn't just a matter of avoiding plagiarism and copyright infringements; it can also be about fostering innovation, creating new products and computer codes, and pushing the boundaries of what GenAI can achieve. Achieving excellence, on the other hand, means ensuring that the output meets or exceeds the standards that would be expected of a human creator by law, ethics, tradition, or industry requirements. Together, these principles form the bedrock of trust and value in GenAI-generated works. This section delves into the critical importance of upholding these standards and the strategies you can employ to achieve them.

Strategies for ensuring originality in GenAI creations

To ensure the originality and quality of GenAI creations, it's essential to implement strategies that can verify the uniqueness and accuracy of the content produced. Following are some strategies you can use to improve content generated by GenAI:

- **Prompt GenAI to provide references in its responses and then review those references yourself.** Why? Because GenAI often lies. And, yes, it can totally make up references to support its lies, too. But also because that's a good first step in spotting hints of plagiarism or copyright infringements.

  

  WARNING That's why you never want to prompt a GenAI model to factcheck or check for plagiarism in its own responses and just accept its answer. Nor should you ever assume that you're using a different model to check the first model's work. You could easily be using two different applications built atop the same GenAI model.
- **Use a different GenAI model to cross-reference generated content against existing works, flagging potential duplicates for review.** For example, prompt Google Smart Search or Perplexity to factcheck the response you got from Claude or ChatGPT. Some GenAI search tools like Perplexity will automatically cite its sources for you to review, too. Make sure to stop and actually review the sources it gives you.
- **Consider embedding digital watermarks or metadata within GenAI-generated content, creating a traceable digital footprint that confirms its origin and authenticity.** See the nearby sidebar, "How to use digital watermarks and embedded metadata," for information on how to do this.

- **Know the data source.** The development of GenAI models should be transparent and ethical, with clear documentation of the data sources and training methods used. By this, we mean that the maker of the GenAI model or application — either a vendor or your internal AI development team — should provide transparent information on how the model was trained and what data was used. This transparency not only builds trust but also allows for the scrutiny necessary to maintain the integrity of models in the rapidly evolving GenAI landscape. Discovering how the model was trained can help you estimate the risks in terms of the likelihood of originality and reliability of outputs.
  For example, the GPT models that power ChatGPT were trained on data scraped from the internet. That may mean a higher risk in the potential of plagiarism and copyright infringement than, say, a specialized model that AI scientists at your company trained only on company data.
- **Consider asking your employer or GenAI vendors to use Causal AI to reveal how the GenAI model came to its decision (output) as a deeper check of its functioning within ethical guidelines.** This is a highly technical strategy that your AI or IT department or GenAI vendor will need to do for you, as it involves developing causal inference algorithms that can work alongside or within GenAI models, creating training datasets that include causal relationship information, designing new model architectures that can combine causal reasoning with natural language generation, and developing metrics to manage the two. However, once in place, the Causal AI can reveal how the GenAI did what it did so you can better evaluate the originality and quality of its outputs.

HOW TO USE DIGITAL WATERMARKS AND EMBEDDED METADATA

Embedding digital watermarks or metadata is a method used to include hidden information within GenAI-generated content to verify its origin and authenticity. This is particularly important for digital works such as images, videos, music, and text that are created by GenAI systems or tools, as it helps to prevent unauthorized use and copyright infringement, as well as to track the distribution of the content.

Digital Watermarks

- **What they are:** Digital watermarks are covert marks or codes that are inserted into digital content. They can be visible or invisible to the human eye (or ear, in the case of audio content). Invisible watermarks are designed to be undetectable under normal circumstances but can be revealed through specific software or techniques.

- **How to use them:** To use digital watermarks, content creators or distributors employ specialized software to embed the watermark into the content before it's distributed. The watermark might contain information such as the creator's identity, the date of creation, or terms of use. If the content is found elsewhere later, the watermark can be extracted to prove its origin or to show that it's been used without permission.
- **Tools for digital watermarks:** Adobe Photoshop is widely used for images, allowing for both visible and invisible watermarking. Software like uMark or Watermarkly can also be used to embed watermarks. For batch processing of images, tools like IrfanView offer plugins for watermarking multiple files at once. For videos, software like Video Watermark Pro can be used. Audio watermarks can be embedded using tools like Audio Watermarking Tools (AWT).

Metadata

- **What it is:** Metadata is data that provides information about other data. In the context of GenAI-generated content, metadata can include details such as the author's name, creation date, location, copyright information, and any other relevant data about the content's origins or rights.
- **How to use it:** Metadata is often attached to files and can be viewed and edited by using various software tools. When generating content, the GenAI system or the user can input metadata into the file properties. For example, in image files, metadata can be embedded into the EXIF data, while for text documents, it might be included in the document properties or within the file itself.
- **For metadata editing:** For images, tools like Adobe Bridge or ExifTool allow users to view and edit EXIF data. For documents, Microsoft Word and Adobe Acrobat can be used to edit properties and metadata. Music files' metadata can be edited with software like MusicBrainz Picard or mp3tag.

Wrap it up and distribute. Verify that the watermark or metadata has been correctly embedded and can be retrieved or viewed using appropriate tools. Then distribute the content however you wish. After the watermark or metadata is embedded, you can distribute the content with the peace of mind that a traceable digital footprint is attached to it.

Maintaining quality standards in GenAI outputs

The foundation of GenAI's output quality lies in the data it's trained on. Using training datasets that are diverse, unbiased, and of high quality helps avoid perpetuating inaccuracies and stereotypes. Additionally, you

should add more current and/or specialized data as needed for the model to make more accurate and timely analyses and responses.



REMEMBER AI models of all types, including GenAI models, age out over time. It's called *model drift* or *model decay,* which refers to a model trained on and working with data that's aged to the point of being irrelevant, inaccurate, untruthful, out-of-date, or otherwise not useful. The effect is that the model has drifted too far from the concept, mission, or data from which it was meant to produce high-quality responses.

Imagine you're a chef who's perfected a recipe your customers love. Over time, the availability and quality of ingredients change, people's taste preferences evolve, and new dietary trends emerge. If you continue to use the same recipe without adapting to these changes, the once-popular dish may no longer satisfy your customers. Similarly, in AI, model drift occurs when the environment around a predictive model changes, making its once-accurate predictions less reliable. Just like the chef needs to tweak the recipe to align with current tastes and ingredient quality, data scientists must update or retrain their AI models to ensure they remain effective as the underlying data and relationships they were built upon shift.

Two main culprits are behind this mix-up: concept drift and data drift. Your recipe's success relies on two key factors: the taste preferences of your customers (concept) and the ingredients you use (data).

- **Concept drift:** Just as customers' taste preferences evolve over time, the underlying patterns and relationships that the GenAI model learned to predict also change. Your recipe must be updated to cater to these new tastes, or it will fail to satisfy your customers, just as the model will fail to make accurate predictions if not updated periodically.
- **Data drift:** Data drift occurs when the input data the model receives is still about the same concept, but the characteristics of the data have changed. The model may then not perform as well because the inputs it was trained on no longer match what it's currently receiving. This is similar to ordering the same ingredients for your prized recipe time and again, but their quality, size, and flavor subtly shift over time and have an adverse effect on your dish.

To ensure your signature dish (or GenAI model) remains a crowd favorite, you must remain vigilant. Regularly assess whether it still delights the palate and confirm that the ingredients (data) you're using are consistent with the recipe as originally crafted. Occasionally, you may need to refine your recipe with fresh, seasonal produce, adjust your seasoning techniques, or even develop an entirely new culinary creation if dining preferences have significantly changed. This could mean retraining your model with new data or starting anew with a different model. That's how you combat model drift and ensure your menu (and predictions) stays delectable, relevant, and satisfying!

But you need to be doing other performance checks, too. GenAI models should be evaluated regularly against quality benchmarks, which can include both automated metrics and human evaluation to gauge the relevance, coherence, and originality of the outputs.



TIP Incorporating human review into the GenAI content-generation process is beneficial. Human experts can offer nuanced feedback and adjustments to refine the GenAI's output, ensuring it adheres to the desired quality standards. Feedback loops are also valuable for improving GenAI models. Typically, that's done by a user clicking on a thumbs-up or thumbs-down icon to indicate to the model when it has hit the mark or missed it in a specific response. But users can give the model feedback in a prompt, too. When you or your company AI team engage in analyzing GenAI performance and integrating user feedback, the model can be iteratively refined for better quality outputs.

Last, but certainly not least, adherence to ethical guidelines and legal standards should be non-negotiable. This means making sure that GenAI responses respect copyright laws and avoid the creation of deceptive or harmful content. It's your or your company's responsibility to manage the quality of outputs and clearly label GenAI-generated content.

Applying Journalism Ethics to GenAI-Generated Content

Applying journalism ethics to GenAI-generated content is a sound and useful way to consistently make sure its responses meet high standards and comply with ethical rules. The following sections provide guidance on approaching GenAI content like a journalist would, which is good advice no matter what field you're in.

Following basic journalistic principles

Even if you aren't a journalist, you can (and should) apply the same following principles when using GenAI to craft content:

- **Transparency:** News media professionals and content providers need to be upfront about using GenAI to generate articles, pictures, podcasts, or anything else. It's critical to let the audience know upfront who (or what) is behind the reporting. If this information comes out after the fact, you'll lose credibility and reader trust.
- **Accountability:** Journalists, content providers, advertising creators, and others have to babysit their GenAI to make sure it's not spouting nonsense or harmful information. Before hitting the publish button, you need to factcheck and edit the content to ensure that the GenAI hasn't made up a story, composed the content badly, left out important details, added untruths, or engaged in any misconduct.
- **Accuracy:** Factchecking and proofreading GenAI content is extremely important. You've got to sift through the GenAI's work to catch any fibs or flubs that could trip up your readers or, worse, cause a ruckus, physical harm, or a lawsuit. Also, keep in mind that if a GenAI model commits to something with your customers or readers, the courts will make you follow through on it.
- **Minimizing harm:** Imagine the consequences if your GenAI starts giving advice. Editors and publishers need to put on their detective hats and verify everything before it goes out, especially when the stakes are high. "Trust but verify" is as important a rule now as it ever was. And that goes for anyone creating any kind of content with GenAI — not just the media.
- **Human oversight:** GenAI may be smart, but it's also dumb as a rock in many ways. It's also restricted in how much reporting it can actually do. It can't, for example, go interview witnesses on the scene, describe the tension in a courtroom at a key moment, or gather data that doesn't yet exist as a digital source. Journalists need to swoop in to get information and deeper insights from undigitized sources and then also factcheck and fix mistakes in GenAI's contribution to the story. The same goes for anyone else making GenAI content. Always, always, always check its homework and add your own expertise, research, and insights as needed.

- **Ethical implications:** Watch out for plagiarism, bias, malice, foul language, and other bad behavior in GenAI-generated content. Always keep in mind that GenAI mimics human behaviors it sees in its data. While it doesn't understand malice, it does learn that people sometimes respond that way, so it mimics that type of response (the same with other emotions and foul language). You, your AI team, or the vendor would be smart to take steps to make the GenAI as well-trained and unbiased as you possibly can.



REMEMBER By sticking to these principles, you can make sure that GenAI-generated content doesn't end up like a bad tabloid headline. Keep it honest, keep it clean, and keep it ethical.

Adhering to truth and integrity in GenAI-assisted reporting

Sticking to the principles of truth and integrity is more crucial than ever when using GenAI tools like ChatGPT for content generation. Keep your moral compass pointing north using journalism values and ethics, whether or not you're working in journalism.

First, remember that honesty is the best policy. When you're using GenAI, be upfront about it. Letting your readers know that part of your content was GenAI-assisted isn't just about being transparent; it's about maintaining the trust that's the bedrock of your relationship with your audience.

Don't just take what GenAI gives you at face value. Put on your detective hat and critically analyze the content. Check the logic, the facts, the math, the graphics, and how it all ties together. Remember, GenAI is a tool, not a replacement for the keen human mind that you bring to the table.



WARNING Steer clear of the plagiarism pitfall. It's tempting to take the GenAI's output and call it your own, but resist that urge. Use GenAI to spark ideas or to get a different angle on a topic, but make sure your final work is genuinely yours.

When GenAI pulls in sources, don't just nod along. Do your due diligence and verify those sources independently. Your reputation for accuracy is on the line every time you choose to publish or share something you generate with GenAI.

Education is your ally. Dive into how your GenAI model of choice works, its strengths, and its weaknesses. The more you know, the better you can use it ethically and effectively. Share this knowledge, too; it's all about lifting the whole field up.

Cite GenAI and/or the sources the GenAI provided for you with the same rigor you would for traditional research. For example, don't just cite Perplexity (a GenAI-powered search engine) in your articles or content; also check the sources it listed in its response, and if those are accurate and reliable, cite those sources as well. It's about setting a standard on par with academic and journalistic standards and showing respect for the information and its origins. You'll appreciate it too when someone else writing a news article or content cites your work specifically and not the search engine. (For example, source: *Artificial Intelligence All-In-One For Dummies* and not "Google AI Overview" or "Perplexity.")

Also, be on guard against GenAI shenanigans when reading or viewing content. You must be just as diligent in guarding against AI shenanigans in emails, direct messages, social media posts, and even phone or video call conversations as people try to present GenAI responses as their own.

Knowledge is power, and in the world of GenAI, it's your superpower. Understand the ins and outs of these tools, from potential biases to ethical use, and you'll be well-equipped to use AI responsibly, enhancing your work without compromising your values or restricting your talent.

Balancing speed with ethical considerations in GenAI content generation

In the fast-moving world of news, journalists are always racing against the clock to get the latest stories out ahead of other news outlets. Assuming you're in a hurry to produce content too, you'll find that with

the help of GenAI tools, there are ways to speed up your work without skidding over ethical speed bumps.

Accuracy, transparency, and accountability shouldn't be sacrificed in the rush. Make it a routine practice to double-check GenAI responses, making sure the facts are straight and the story's told fair and square. It's about keeping it real and reliable, even when the GenAI's doing some or most of the heavy lifting.

To keep everything up and up, set well-defined rules for GenAI use within your company, team, or for yourself. Establish checkpoints where content is to be reviewed by human eyes to give everything the once-over and to (hopefully) catch problems like plagiarism, copyright infringements, biases, or slip-ups. Whatever processes and policies you come up with, keep in mind that the goal is to find that sweet spot where getting the news out quickly doesn't mean cutting corners on the accuracy or responsibilities.



REMEMBER By balancing the need for speed with a commitment to ethics, you're steering GenAI's power in the right direction. You're proving that you can keep up with the times while staying true to the heart of what you know and believe to be right: telling it like it is, sticking to the facts, and keeping the trust of the people tuning in.

Joining the Responsible AI Movement

As GenAI tools become more integrated into our daily lives, the importance of steering these technologies in a direction that safeguards human dignity, rights, and freedoms can't be overstated. Joining the responsible AI movement helps everyone push toward a future in which technology serves the greater good and aligns with collective values and ethical principles. The following sections cover the basics of this movement and ways that you can get involved.

Understanding the goals of the responsible AI movement

Responsible AI ensures that the AI models — not just GenAI models — humans create align with societal values and operate within ethical boundaries. This means designing AI models that respect privacy, promote fairness, and are transparent in their operations. The policies and processes governing AI model training, retraining, and uses are important parts of responsible AI, too. By joining the responsible AI movement, individuals and organizations commit to developing AI models that are not only efficient but also equitable and accountable.

The movement emphasizes the importance of inclusivity in AI development. Diverse teams are more likely to identify potential biases in AI systems, leading to more representative and unbiased outcomes. This inclusivity extends to the data used to train AI, ensuring it reflects the diversity of the real world and mitigates the risk of perpetuating existing inequalities.

The responsible AI movement also advocates for the development of AI that is secure and resilient. As AI systems become more complex, the potential for exploitation by malicious actors increases. By adhering to best practices in AI security, you can protect sensitive data and critical infrastructure from cyber threats.

Moreover, the movement encourages ongoing dialogue between technologists, policymakers, and the public. This collaboration is essential for creating a regulatory environment that fosters innovation while protecting the public interest. It also helps demystify AI, making it more accessible and understandable to non-experts, which is crucial for informed public discourse.

Finally, joining the responsible AI movement is a commitment to continuous learning and improvement. AI is a rapidly evolving field, and what is considered responsible today may change tomorrow. Being part of the movement means staying informed about the latest developments, challenges, and opportunities in AI ethics. It also means you have the opportunity to add your voice and thoughts to the evolution of AI and its uses.



REMEMBER The responsible AI movement is not just about preventing harm; it's about proactively contributing to a future in which AI enhances our capabilities, complements our humanity, and upholds our shared ethical standards. It's a collective effort to ensure that as AI's role in society grows, it does so in a way that's beneficial for all.

Contributing to ethical AI development and use

Ethical AI development and use requires the collective effort of many to ensure it grows healthily and sustainably. Everyone, from tech developers to end-users (yes, this means you!), has a role to play in cultivating an environment where AI operates within the bounds of ethical principles:

- **For tech developers and data scientists,** the journey begins with an awareness of the issues so that work can be done to effectively address them when training, retraining, fine-tuning, or otherwise adapting AI models. Understanding the ethical implications of AI is paramount to these efforts. The implications include recognizing the potential for bias in datasets and algorithms and the consequences these biases can have on society. Developers must strive to create inclusive AI by ensuring diversity in training data and testing AI systems across a wide range of scenarios to identify and mitigate biases.
- **Business leaders and policymakers** also have a crucial role. They must establish and enforce ethical guidelines for AI development and use within their organizations and at a legislative level. This role includes creating transparent policies around data usage, privacy, and consent, as well as setting up oversight committees to monitor AI systems for ethical compliance.
- **End-users,** on the other hand, can contribute by staying informed and demanding transparency. By understanding how AI systems make decisions and the data they use, users can hold companies accountable for the AI products they offer. Public discourse and advocacy for ethical AI can influence businesses and governments to prioritize ethical considerations in their AI initiatives.

  

  TIP Start by making companies do more than give lip service to Responsible AI. Specifically, require them to disclose what and how they're implementing Responsible AI practices in their real-world AI deployments.
- **Ethicists, sociologists, legal experts, and technologists** must work together to foresee the societal impacts of AI and guide its development accordingly. This

collaborative approach ensures that multiple perspectives are considered, leading to more robust ethical frameworks.

- **Educational institutions** have a responsibility to integrate ethics into STEM curricula, preparing the next generation of AI professionals to approach their work with a strong ethical foundation. Continuous professional development in ethics should also be encouraged within the tech industry.
- **Investors and shareholders** can influence ethical AI by supporting companies that prioritize responsible AI practices. By directing funds toward ethical AI initiatives, they can incentivize companies to adopt best practices and contribute to the development of AI that benefits society.
- **The media** play a pivotal role in shaping public perception of AI. Responsible reporting on AI should highlight both its potential benefits and ethical concerns. This balanced approach can foster a more nuanced understanding of AI among the general public. It can also prepare the public to reject AI-produced propaganda, deepfake videos, political and societal manipulations, and AI-produced conspiracy theories and fake data.

Aligning with global efforts for responsible AI practices

Several global efforts are aimed at developing, tracking, coordinating, or reporting on responsible AI practices. These efforts tend to be massive, as they involve a multi-layered approach that includes individual action, organizational commitment, and international collaboration. Following are some examples of global efforts currently underway to promote and implement responsible AI, led by various international organizations, governments, and institutions:

- **UNESCO's Global Index on Responsible AI:** This initiative aims to monitor the implementation of globally established human rights-based AI principles. The index seeks to measure how well countries are implementing practices that ensure AI systems are transparent, accountable, and aligned with human rights and democratic values. Find more information here: `https://www.unesco.org/en/articles/tracking-national-commitments-global-index-responsible-ai`
- **Atlantic Council's GeoTech Center:** The Atlantic Council's GeoTech Center is an initiative focused on the intersection of geopolitics, technology, and governance. As part of its broader mission, the GeoTech Center has a dedicated effort towards responsible AI, which aims to ensure that artificial intelligence and emerging technologies are developed and implemented in ways that are ethical, beneficial for society, and promote international security and prosperity. Find

more information here: https://www.atlanticcouncil.org/programs/geotech-center/advancing-responsible-ai-globally/

- **U.S. Government Initiatives:** The United States is leading global efforts to build strong norms that promote the responsible military use of AI. This includes endorsing responsible AI measures for global militaries. Examples include:
  - **National AI Initiative (NAII):** The National AI Initiative Act of 2020 aims to coordinate a national strategy on AI across various sectors, including international cooperation and coordination. It emphasizes the importance of leadership in AI ethics and governance on the global stage.
  - **The AI Partnership for Defense:** Led by the DoD's JAIC, this initiative involves partnership with military allies and partners to share best practices, collaborate on ethical AI frameworks, and ensure that the use of AI in defense aligns with shared democratic values and international humanitarian law.
  - **Global Partnership on Artificial Intelligence (GPAI):** The United States is a founding member of GPAI, an international initiative that brings together experts from industry, government, civil society, and academia to advance the responsible development and use of AI, guided by human rights, inclusion, diversity, innovation, and economic growth.
  - **USAID's Digital Strategy:** The U.S. Agency for International Development (USAID) Digital Strategy includes efforts to leverage AI and other digital technologies in international development. It focuses on the responsible use of AI to improve program outcomes while mitigating risks, such as bias and threats to privacy.
  - **Engagement in International Standards Organizations:** The United States actively participates in international standards organizations, such as the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), contributing to the development of global standards for AI that incorporate principles of responsible AI.
- **World Economic Forum's AI Governance Alliance:** This initiative brings together global stakeholders to champion the responsible design and release of transparent and inclusive AI systems. You can find more information here: https://initiatives.weforum.org/ai-governance-alliance/home
- **Responsible AI Institute (RAI Institute):** This global, member-driven nonprofit organization is dedicated to enabling successful, responsible AI efforts in organizations. RAI Institute provides independent assessments and certifications for AI systems, aligning them with existing and emerging policies, regulations, laws, and best practices. Find more information here: https://www.responsible.ai/
- **AI for Good:** Led by the International Telecommunication Union (ITU), AI for Good is a movement and a series of global summits that aim to ensure that AI benefits humanity and contributes to the United Nations' Sustainable

Development Goals. Find more information here: https://aiforgood.itu.int/

- **Organization for Economic Cooperation and Development (OECD):** The OECD has developed AI Principles that have been adopted by many countries. These principles promote AI that is innovative, trustworthy, and respects human rights and democratic values. Find more information here: https://www.oecd.org/en/topics/policy-issues/artificial-intelligence.html
- **Global Partnership on Artificial Intelligence (GPAI):** GPAI is an international initiative to support responsible and human-centric development and use of AI, guided by human rights, inclusion, diversity, innovation, and economic growth. Find more information here: https://gpai.ai/
- **Partnership on AI (PAI):** PAI is a coalition that includes major tech companies, civil society groups, media organizations, and academic institutions. The Partnership on AI works to study and formulate best practices on AI technologies and to advance the public's understanding of AI. Find more information here: https://partnershiponai.org/
- **IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems:** The IEEE Standards Association has launched a global initiative that has created a comprehensive set of guidelines and standards to promote ethical considerations in AI and autonomous systems. Find more information here: https://standards.ieee.org/industry-connections/ec/autonomous-systems/

There are also several national and regional efforts with global impact. Many countries have developed their own frameworks and guidelines for AI ethics. For example, Singapore's Model AI Governance Framework, China's New Generation Artificial Intelligence Governance Principles, AI Ethics Guidelines from the European Union (EU), and the United States' National AI Initiative.



TIP We've included URLs in the previous list in case you want to take a deeper look at the resources, research, and updates that are included on their websites. Additionally, academic journals, conferences, and workshops frequently address the topic, offering a platform for experts to discuss and advance the field. If you can't make it to the conferences, look for YouTube videos and news reports to glean some of the information that was shared there. Responsible AI efforts and guidelines are rapidly

evolving, so staying informed through these organizations and their publications is a good way to keep up with the latest developments.